

Introduction to Gravitational Waves and Pulsar Timing Array



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Chapter 1

Introduction to Pulsars

What are pulsars?

Pulsars are magnetised neutron stars that appear to emit periodic short pulses of radio radiation (electromagnetic waves at the lowest-frequency, longest-wavelength end of the electromagnetic spectrum).

The time period of their oscillations is between 1.4 ms and 8.5 s. A neutron star is one with a very small radius and an extremely high mass. As a result, the density of one is very high.

In the event of such high pressure as a supernova, the protons and electrons of the atoms involved undergo a phenomenon called inverse beta decay.

Inverse beta decay:

$$p + e^- \longrightarrow n + \nu_e \quad (1.1)$$

Pulsars are not literally ‘pulsating’. The neutron stars that rotate (at very high speeds) emit beams of radiation, which are detected by the observer, and cause a **lighthouse effect**. Even though the mechanism of this radiation is not well known, it has various applications.

Applications of neutron stars and pulsars include **using it to study extreme conditions of matter** (The extremely high gravitational potential, density (more than that of atomic nucleus), and magnetic field strength), and **using it as a clock** (Pulse periods are very accurate. So much so that it permits sensitive measurement of precision relativistic observables).

1.1 Neutron star masses and densities

What really causes these pulses though? It is the combination of various factors.

Rapid rotation of the neutron star with a strong magnetic field. A misalignment between the magnetic axis and rotation axis.

For a pulsating star, density controls the pulsation period. The gravity pulls the star inwards, while the pressure pushes it out. By this, the speed of oscillation depends on the the gravity per unit area.

Higher density $\rightarrow$ strong gravity $\rightarrow$ faster oscillations

Lower density $\rightarrow$ weaker gravity $\rightarrow$ slower oscillations

The period of oscillation for sun is in hours or days. But the observed period of the pulsars is in the range of ms to s. So the speed of oscillations has to be much faster, which means the star has to be of higher density. The neutron star fits perfectly in the calculations (it is over two hundred trillion times denser than the sun).

Mathematically, the condition that we can apply here, considering how density controls the period of oscillation: The centrifugal acceleration at its equator does not exceed the gravitational acceleration.

$$F_{centripetal} < F_{gravitational}$$

$$(acc^n)_{centripetal} < (acc^n)_{gravitational}$$

$$\omega^2 R < \frac{GM}{R^2}$$

Substitute and write the angular acceleration ' ω ' in terms of 'T'

$$\left(\frac{2\pi}{T}\right)^2 R < \frac{GM}{R^2}$$

Simplifying the equations further,

$$T^2 > \left(\frac{4\pi^2}{GM}\right) R^3$$

$$T^2 > \left(\frac{4\pi R^3}{3}\right) \frac{3\pi}{GM}$$

Writing this in terms of mean density (ρ) ie. $\frac{M}{\frac{4}{3}\pi R^3}$

$$\frac{M}{\left(\frac{4}{3}\pi R^3\right)} > \frac{3\pi}{GT^2}$$

$$\rho > \frac{3\pi}{GT^2}$$

$$T = \left(\frac{3\pi}{G\rho}\right)^{1/2} \quad (1.2)$$

Chandrasekhar Mass:

A dense star is kept intact due to the presence of the balancing forces, **the gravity** (tries to collapse the star) and **the electron degeneracy pressure** (the pressure coming due to the intense packing of the electrons)

As the mass of star increases, the electron degeneracy pressure no longer stays enough to be equal to the gravitational attraction. This limit is called the **Chandrasekhar limit**.

$$M_{Ch} = 1.4M_o \quad (1.3)$$

Substitute the values for the boundary condition of mass equal to 1.4 times that of sun. The first ever neutron star to be discovered had a time period of 1.4×10^{-3} seconds. For such a star, the radius comes out to be equal to around 20 km. Thats it!

This is how dense these objects really are..!

Canonical Neutron stars:

These are objects with a mass of around 1.4 times that of the sun, and radius of approximately 10 km. The moment of inertia of such an object (given solid sphere) will be:

$$I = \frac{2}{5}MR^2$$

for R = Radius of the sun and $M = 1.4$ times mass of the sun.

$$I = \frac{2}{5}(1.988 \times 10^{30})(1.4 \times 696342)^2$$

$$I = 0.76 \times 10^{41} kgm^2 \quad (1.4)$$

The Chandrashekhar limit is the lower limit for the existence of neutron stars. The upper limit can be said equivalent to 3 times the mass of sun. If the upper limit is crossed, the neutron star must collapse into a black hole.

Equation of States (EoS):

The equation of states tell us how the pressure is dependent on the density. Gravity tries to collapse the star while internal pressure resists collapse. The EoS determines how much mass can be supported.

$$1.4M_o < M < 3M_o \quad (1.5)$$

1.2 Magnetic field and Dipole Radiation

Magnetic Fields:

The stars are said to have an approximate magnetic dipole (every massive object does, take the example of our Earth. Magnetic dipoles exist, and so there are magnetic field lines present around the planet).

Now we are talking of compressing this massive star into a small object (say $10^6 km$ to $10 km$ radius), the cross sectional area reduces by 10^{10} times ($\propto r^2$). So a young neutron star shall have massive dipolar fields. Conservation of angular momentum also confirms that, the smaller the radius, the faster the spin.

Magnetic Dipole Radiation:

Say, the magnetic dipole moment is perfectly aligned with the rotation axis of the neutron star. Then the magnetic field pattern remains the same, and there is no emission of any electromagnetic radiation.

This is an imaginary case. Let that get tilted by an angle α . Larmor's equation: Any charged particle radiates when accelerated and that the total radiated power is proportional to the square of the acceleration. Mathematically,

$$P = \frac{2}{3} \frac{q^2 \dot{v}^2}{c^3} \quad (1.6)$$

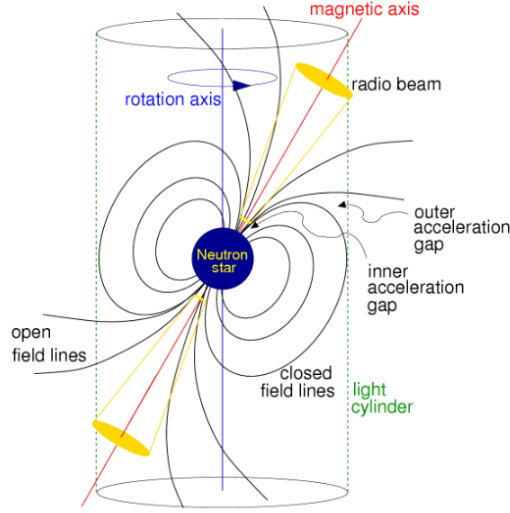


Figure 1.1: Magnetic field lines around a neutron star

Rewriting the Larmor's formula in terms of rotating electric dipole,

$$P = \frac{2}{3} \frac{(q\ddot{r} \sin\alpha)^2}{c^3}$$

$$P = \frac{2}{3} \frac{\ddot{p}_\perp^2}{c^3} \quad (1.7)$$

where $m_\perp = m \sin\alpha$ is the perpendicular component of the magnetic dipole moment.

1.3 Spin-Down Luminosity

The term 'spin down' refers to the neutron star **losing** energy as it spins. This indicates the amount of power the pulsar is releasing into its surroundings. It is the **rate of slowing down of the star**.

The rotational kinetic energy E of a spinning object is given by

$$E = \frac{1}{2} I \omega^2$$

$$E = \frac{I}{2} \left(\frac{2\pi}{T} \right)^2$$

$$E = \frac{2I\pi^2}{T^2} \quad (1.8)$$

The moment of inertia of a solid spherical object of mass M and radius R is given by

$$I = \frac{2}{5} MR^2$$

Now from equation (1.4),

$$I = 0.76 \times 10^{41} \text{kgm}^2$$

We plug in the values for I and $T = 0.033\text{s}$ of a crab pulsar.

$$E = 1.8 \times 10^{49} \text{ergs} \quad (1.9)$$

Now we try to determine the rate at which the neutron star is losing energy. Moving to equation (1.8),

$$E = \frac{1}{2} I \omega^2$$

$$-\dot{E} = -\frac{d}{dt} \left(\frac{1}{2} I \omega^2 \right) = -I \omega \dot{\omega}$$

This quantity $-\dot{E}$ is called the **spin down luminosity**.

Substitute $\omega = \frac{2\pi}{T}$

$$-\dot{E} = -I \left(\frac{2\pi}{T} \right) \left(\frac{\dot{2\pi}}{T} \right)$$

$$-\dot{E} = \frac{4\pi^2 I}{T^3} \dot{T} \quad (1.10)$$

The Crab pulsar has $T = 0.033\text{s}$ and $\dot{T} = 10^{12.4}$. If $I = 10^{45} \text{gcm}^2$, the neutron star is losing rotational energy at that rate:

$$-\dot{E} = 4 \times 10^{38} \text{ergs}^{-1} \quad (1.11)$$

Minimum Magnetic Field Strength:

We have so far seen, that the neutron star loses energy through magnetic dipole radiation. There should be a minimum magnetic field originally, below which the star cannot afford to lose its field anymore. This minimum is the minimum magnetic field strength of that star.

Mathematically,

$$P_{rad} = -\dot{E}$$

Substituting the values for each of the sides,

$$\frac{2}{3c^3} (BR^3 \sin\alpha)^2 \left(\frac{4\pi^2}{P^2} \right)^2 = \frac{4\pi^2 I \dot{P}}{P^3}$$

$$B = \left(\frac{3c^3 I}{8\pi^2 r^6} \right)^{1/2} \frac{\sqrt{P \dot{P}}}{\sin\alpha}$$

Now, we know,

$$0 \leq \sin\alpha \leq 1,$$

$$\frac{1}{\sin\alpha} \geq 1$$

So,

$$B \geq \left(\frac{3c^3 I}{8\pi^2 r^6} \right)^{1/2} \sqrt{P\dot{P}}$$

which represents the minimum possible magnetic field.

When we plug in the values for a typical neutron star, we get the minimum magnetic field strength at the surface of a canonical pulsar,

$$B \geq 3.2 \times 10^{19} \sqrt{P\dot{P}}$$

(in CGS unit gauss)

For the **Crab pulsar**, the minimum magnetic strength is

$$\left(\frac{B}{\text{gauss}} \right) \geq 4 \times 10^{12} \quad (1.12)$$

Characteristic Age:

Since $P_{rad} = -\dot{E}$, we go further and make assumptions about the Pulsar's age τ , subject to one condition. The pulsar's initial period P_o is much shorter than its current period P (we have approximated $(P^2 - P_o^2 = P^2)$ in the proof).

$$\tau \equiv \frac{P}{2\dot{P}} \quad (1.13)$$

For the Crab Pulsar, $\tau \approx 1240$ yrs, and it will continue to spin for millions of years.

Note that the equation (1.13) works on the approximation about the periods of the pulsars. As of 2026, the actual age of the Crab Pulsar is 972 years (since its supernova was observed in 1054 AD).

1.4 Braking Index

The braking index describes how the rotation of a pulsar slows down with time. It defines the proportionality relation between the rate of change of angular rotation frequency and the angular rotation frequency itself, ie.

$$\begin{aligned} \dot{\omega} &\propto -\omega^n \\ \dot{\omega} &= -K\omega^n \end{aligned} \quad (1.14)$$

where n here is the Braking Index.

Physically, n here is the opposing torque, that is slowing down the pulsar. We can very well see how the **braking index is closely linked with the Spin-down luminosity**.

For the Magnetic dipole radiation, the braking index is expected to be ≈ 3 .

$$\dot{\omega} = -K\omega^3$$

Now take the equation (1.14)

Differentiate both sides wrt time.

$$\ddot{\omega} = -K \left(n \omega^{n-1} \right) \dot{\omega}$$

$$\ddot{\omega} = -K \left(n \frac{\omega^n}{\omega} \right) \dot{\omega}$$

$$\ddot{\omega} = n \frac{\dot{\omega}^2}{\omega}$$

Rearranging,

$$n = \frac{\omega \ddot{\omega}}{\dot{\omega}^2} \quad (1.15)$$

The observed braking index values for pulsars are $1.4 \leq n < 3$.

1.5 The Life of a Pulsar

Let's look at the $P - \dot{P}$ Diagram, which is a visual representation of the entire life journey of the pulsar.

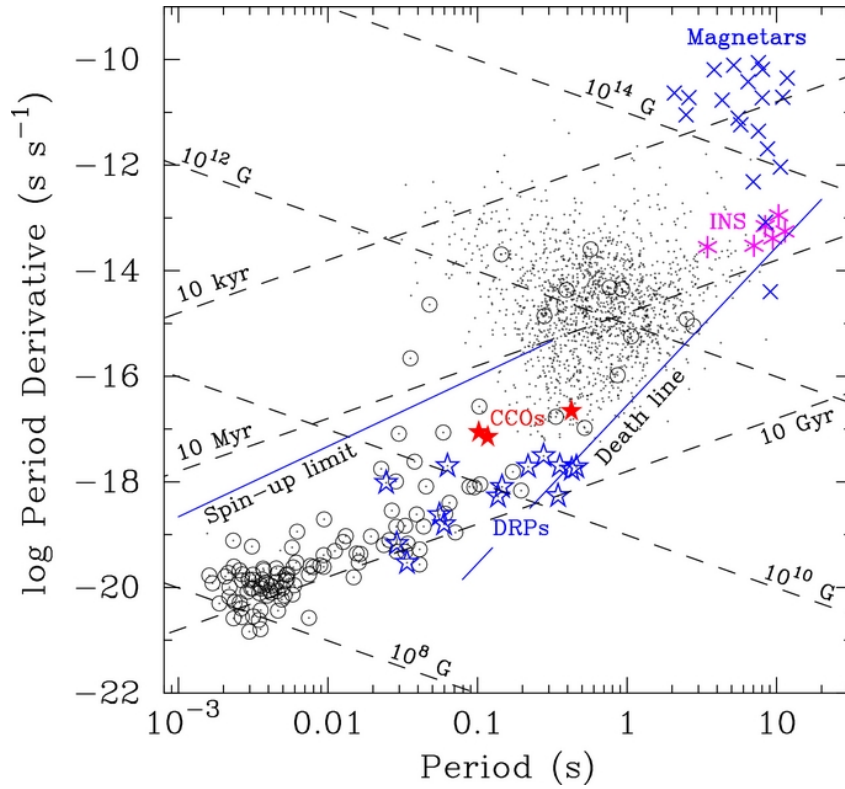


Figure 1.2: $P - \dot{P}$ Diagram of Pulsars

The spin period vs spin period derivative (how quickly the pulsars spin rate is slowing due to loss of luminous energy) diagram shows the different classes of neutron stars. From it, we have understood different properties of the neutrons stars, how they change over time, etc.

The life of pulsars is shown by the dotted line (with negative slope) in the above figure, starting at the top left corner and going towards the bottom right corner as its age goes on increasing. We can see the actual values as this happens (in CGS units Gauss).

The linear graph shows that as the characteristic age increases, the period of rotation also increases. It is understandable - a newly born neutron star has a very high angular momentum, so the pulsar 'pulsates' very fast. As the time goes on, the period of 'pulsation' goes up.

Also, the rate of increase of period (shown on the y-axis) is negative, which accounts for the energy reducing as the characteristic age of the star increases.

As the core collapses, electrons start combining with protons, forming neutrons and neutrinos, as seen previously in eq. (1.1). The density of the core increases rapidly, until it reaches that of a atomic nuclei. At this point, the core rebounds ie. produces a shock wave moving outward through the star. This is nothing but a **Supernova explosion**. What is left in the center is the neutron star (given it does not form a black hole), which is often observable due to the pulsars.

Interestingly, the pulsars with a characteristic age less than certain limit stay in the supernova remnants itself.

Rotating Radio Transients (RRATs):

Pulses from a pulsar are supposed to be very uniform. Their applications involve using it as clocks. But then there comes this species of pulsars called RRAT's. These, in a way, pulsate randomly.

These pulsars are observed to have $P > 1$ second, which means that it will lie in the right hand side of the graph. It is thought that these might be very old pulsars, almost close to turning off.

Recycled Pulsars:

Sometimes the binary system survives the supernova. In that case the neutron star keeps orbiting another star (the companion). The companion can be a normal star, or another neutron star. Now say, the magnetic field is very weak for the neutron star. It will get matter from the companion, and hence have enough energy to emit radiation, forming a pulsar.

These are **millisecond pulsars**, meaning pulsars with period in milliseconds. These were not millisecond pulsars when they were born, but rather normal pulsars spun up by the accretion of the binary companion. These are rotationally more stable, and hence can act as good clocks.

1.6 Emission Mechanisms

Near the pulsar, there is a region filled with

- strong magnetic fields
- electrons and positrons (charged particles)
- intense electromagnetic radiation

This region is called **Magnetosphere**. The radio pulses originate in the pulsar magnetosphere. The neutron star is a dipole, whose magnetic axis is tilted at an angle to the rotation axis (it very well acts like a bar magnet spinning along an axis tilted to a small angle).

As the magnetic fields rotate, they induce an electric field. This is in accordance with Faraday's law. The neutron star acts as a **unipolar generator**, i.e., a conducting object that produces an electric field. Analogous to the bar magnet, the neutron star has two

poles at opposing ends. This generates a potential difference (voltage) between different parts.

The total **Lorentz force** acting on a charged particle is given by

$$\vec{F} = q \left(\vec{E} + \frac{\vec{v} \times \vec{B}}{c} \right) \quad (1.16)$$

Near the equatorial region, the electric charges redistribute themselves such that the electric field cancels the magnetic field, so $\vec{F} = 0$. The magnetic field lines emerging from the polar caps are supposed to close. However, the ones that cross the **light cylinder** (the distance where the corotation speed would equal the speed of light, i.e. the distance where the magnetosphere extends) do not form a closed loop as expected. This is because then the outer part of the line would have to travel faster than the speed of light, which is physically not possible.

Electrons in the polar cap region are accelerated to very high energies along the closed magnetic field (Any accelerating charge emits electromagnetic radiation. Even if the speed stays constant, a particle is accelerating if its direction changes.). These get accelerated to such relativistic velocities that they start emitting gamma rays, strong enough to interact with magnetic field lines and start producing electron-positron pairs. Now these pairs again interact with magnetic fields, and produce new gamma rays. And these gamma rays again interact to produce new electron-positron pairs, and so on. This makes them emit **curvature radiation** that is strongly polarised in the plane of curvature.

The pulsars 'die' as they lose enough magnetic field strength, and start having much higher time peroids. Graphically, the death is represented at the bottom right corner.

Chapter 2

Pulsars and the Interstellar Medium

A pulsar is a highly magnetized rotating Neutron Star. We get the radiations from these objects in the form of radio waves. The radio waves travel through the interstellar medium (ISM), which is ionized (contains weak magnetic fields). Radio waves are electromagnetic (EM) waves. In the influence of the weak magnetic fields, its plane of polarization will rotate, something known as **Faraday's effect**.

The refractive index for electromagnetic waves in a plasma is

$$\mu = \sqrt{1 - \frac{\nu_p^2}{\nu^2}} \quad (2.1)$$

μ is the refractive index of the EM wave

ν is the frequency of the radio wave

ν_p is the plasma frequency ($\sim 1.5 \text{ kHz}$ for a typical ISM)

A special case arises when $\nu_p > \nu$. In that case the term inside the square root becomes negative, i.e. μ becomes imaginary. So, it is said that the radio wave simply cannot travel through the plasma.

For propagating radio waves, $\mu < 1$. The **Group Velocity** i.e. the collection of many waves with slightly different frequencies is given by

$$v_g = \mu c$$

For most radio observations, $\nu \gg \nu_p$. So we can apply binomial expansion on (2.1), and expand the term for group velocity as

$$v_g \approx c \left(1 - \frac{\nu_p^2}{2\nu^2} \right) \quad (2.2)$$

2.1 Dispersion delay

Dispersion delay is the extra travel time caused by the plasma comparison with travel in a vacuum.

The **Radio waves** are a collection of beams that slightly vary in frequency. The ISM contains free electrons. So there is a difference in the time required by the high-frequency waves (faster) and low-frequency waves (slower). This time difference is called dispersion delay.

Time taken by the wave to travel small distance ' dl ' in plasma:

$$dt = \frac{dl}{v_g}$$

where v_g is the group velocity.

$$t = \int_0^d \frac{dl}{v_g}$$

Travel time in vacuum:

$$t_{vac} = \int_0^d \frac{dl}{c} = \frac{d}{c}$$

Dispersion delay is given by:

$$\int_0^d \frac{dl}{v_g} - \int_0^d \frac{dl}{c}$$

$$t = \int_0^d \left(\frac{1}{v_g} - \frac{1}{c} \right) dl$$

Substitute value for v_g from equation (2.2),

$$t = \frac{1}{c} \int_0^d \left(1 + \frac{\nu_p^2}{2\nu^2} \right) dl - d \quad (2.3)$$

Substitute the values of the constants,

$$\left(\frac{t}{sec} \right) = 4.149 \times 10^3 \left(\frac{DM}{pc.cm^{-3}} \right) \left(\frac{1}{(\nu/MHz)^2} \right) \quad (2.4)$$

where DM (Dispersion measure) is the total number of free electrons along the line the pulsar is being observed.

It is the total number of electrons that the radio wave has to travel through to reach the observer.

$$DM = \int_0^d n_e(l) dl$$

$n_e(l)$ is the electron density at a distance l (in electrons/ cm^3)

dl is the small segment of distance (in parsec (pc))

d is the total distance between observer and pulsar (in parsec (pc))

Now let us try plotting a graph for frequency (in Hz) on the y-axis and pulse phase (rotation in form of turns, written as 0 to 1) on the x-axis.

We see from the graph that the bright track is slanted. It represents the frequency dependent delay. From equation (2.4), we clearly see $t = 1/\nu^2$.

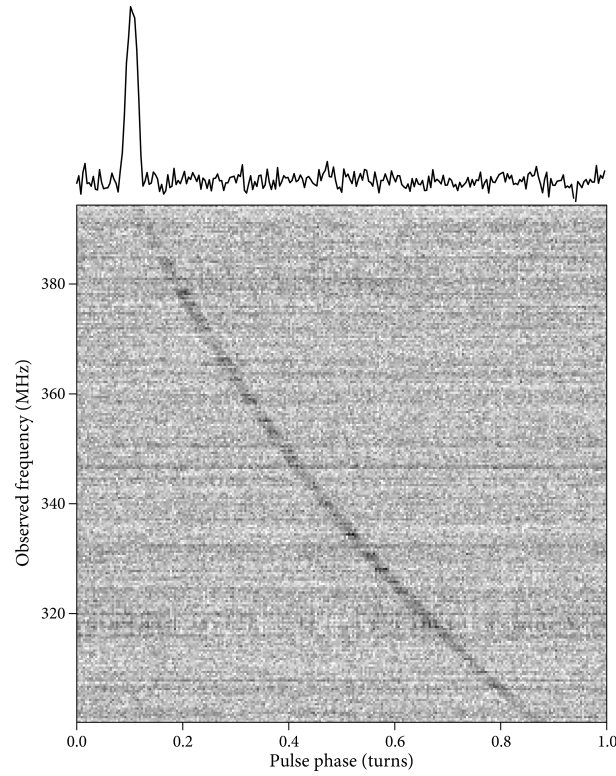


Figure 2.1: [Pulsar Dispersion](#)

In an ionized medium such as the ISM, lower frequencies travel slower than the higher ones. As a result, the pulse gets spread out in time, and is difficult to detect. This is called **Dispersive smearing**. Now, this is a problem. The best method useful for resolving such an issue is **Coherent dedispersion**. It works by multiplying by the inverse filter. This method removes delays as well as smearing

2.2 Scatter Broadening

A phenomenon related to scintillation (twinkling effect due to atmosphere) is **Pulse broadening due to scattering** of radio waves. Due to the ISM, different frequencies travel at different speeds. Some rays travel longer distances because they do not follow a straight line path.

From the figure that follows, we make the following observations:

- The pulse is sharp at high frequencies.
- At low frequencies, it becomes wide, asymmetric and develops a long tail.

It is said to reduce the observed flux density. Note that this does not mean that it destroys energy; it only redistributes it in time.

It becomes more difficult to determine the exact peak position. Also, it makes it difficult to obtain the true time of arrival of the pulse. The time is displayed between two vertical dotted lines in each scatter plot. Note how it increases and becomes more pronounced uncertain.

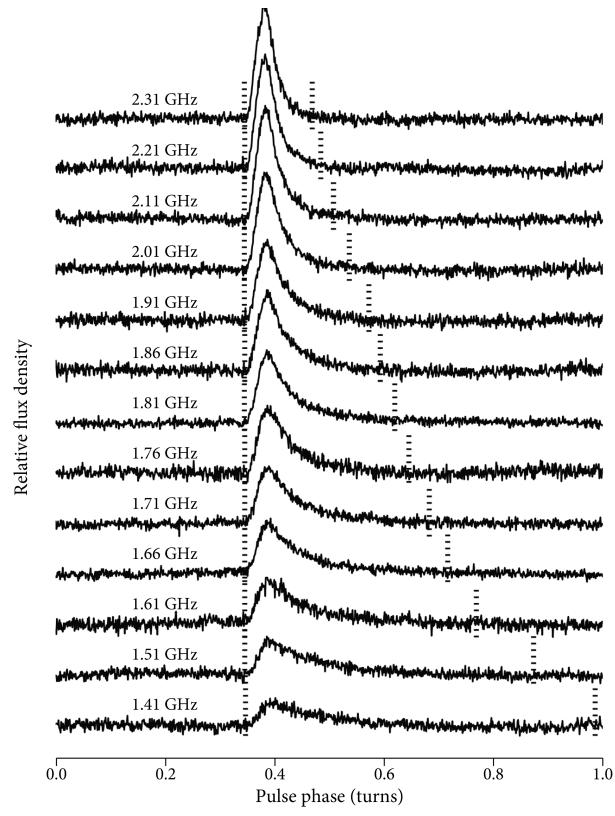


Figure 2.2: [Pulse broadening caused by scattering](#)

Chapter 3

Pulsar Timing

Now we move on towards the most interesting part - the actual use, why one should study pulsars. For pulsar timing, astronomers fold (take average) the data of the instantaneous pulse period P . The instantaneous pulse is of frequency $f = 1/P$, instantaneous pulse phase $\phi = \int f dt$ ($0 < \phi < 1$).

Averaging over many pulses gives an **Average pulse profile**. This adjusts noise.

We define **offset** as a measurable shift between two things. Here, phase offset is the measure of how much the observed pulse is shifted relative to template (ideal pulse profile).

Basically, $\Delta t = \text{Average pulse profile} \times \text{Instantaneous pulse period}$.

Time offset is considered to create **Time of arrival**.

$$\Delta t = \Delta \phi \times P$$

Time of arrival is this time offset added with the reference time.

(TOA is absolute time when the pulse arrives. Time offset is the relative shift)

3.1 Precision in Pulsar Timing

The precision with which TOA is measured (σ) is inversely proportional with **the signal to noise** (S/N) ratio of averaging profile.

$$\begin{aligned}\sigma_{TOA} &\propto \frac{W}{(S/N)} \\ \sigma_{TOA} &\approx \frac{W}{(S/N)}\end{aligned}\tag{3.1}$$

since proportionality constant ~ 1 .

where W is the effective pulse width. Smaller the value of W , sharper and narrower the pulse; and better the precision.

With what precision can timing determine the spin frequency f of a pulsar? The precision is based on how precisely a change in phase $\Delta \phi$ can be measured over some time interval ΔT from the relation $\Delta f = \frac{\Delta \phi}{\Delta T}$.

Now we take an example of the original millisecond pulsar B1937 + 21.

The time period P for the taken example is measured to be ≈ 0.00156 s.

Calculating,

$$\Delta f = \frac{\Delta \phi}{\Delta T} = \frac{6 \times 10^{-4}}{7.5 \times 10^8 \text{ s}^{-1}} = 8 \times 10^{-13} \text{ Hz}$$

$$f = \frac{1}{P} = \frac{1}{0.00156 \text{ s}} \approx 641 \text{ Hz}$$

So the absolute error is:

$$\frac{\Delta f}{f} = \frac{8 \times 10^{-13} \text{ Hz}}{641 \text{ Hz}} = 1.248 \times 10^{-15} \quad (3.2)$$

which is known to 15 significant figures!

Many **Correction factors** have to be checked before we proceed with our study of Pulsar timing. Precisely, the following equation takes all these factors into account.

$$t = t_t - t_o + \Delta_{clock} - \Delta_{DM} + \Delta_{R\odot} + \Delta_{E\odot} + \Delta_{S\odot} + \Delta_R + \Delta_E + \Delta_S \quad (3.3)$$

t_t (**topocentric time**) is the time at which the pulse is received at the telescope on Earth.

t_o (**reference time**) is a fixed reference time chosen as a starting point. It is an arbitrary point; but is fixed in the start.

So essentially,

$$t = t_t - t_o + \text{corrections}$$

Let us look at each of these delays in detail:

- **Clock correction** (Δ_{clock}):
The telescopes are not perfect. This correction is required to correct the differences between the observatory clock and the standard (say UTC time).
- **Correction due to Dispersion medium** (Δ_{DM}):
The Radio waves travel through the ISM. We have previously seen the effects this causes. This needs to be corrected.
- **Roemer delay** ($\Delta_{R\odot}$ and Δ_R):
The earth orbits the sun. The relative distance between the pulsar and the instantaneous position of earth changes. This accounts for the Roemer delay ($\Delta_{R\odot}$). Similarly, consider a binary system of neutron stars rotating along their center of mass. A similar effect will be noticed in this case too (Δ_R).
- **Einstein delay** ($\Delta_{E\odot}$ and Δ_E):
According to Einstein's special relativity, phenomena such as time dilation near massive objects are observed. The Sun's gravity causes redshifting as Earth revolves around it. Clocks run differently around these massive objects. Thus, there is a need for a correction factor (accounts for $\Delta_{E\odot}$).
Similar effects are visible in a binary star system as well (accounts for Δ_E)
- **Shapiro delay** ($\Delta_{S\odot}$ and Δ_S):
The Shapiro delay ($\Delta_{S\odot}$) is a result of curved space-time. The path near the massive is curved as a result of the gravity. So the it takes longer time to move along it. A similar delay (Δ_S) is observed when the space-time is curved along the binary star system. Again, it is same as Solar System Shapiro delay ($\Delta_{S\odot}$), but inside the binary.

To conclude, the errors are caused as a result of imperfections in the instrumentation, radio-wave propagation through the ISM, geometry (inside the solar system, or inside the binary), and relativity (inside the solar system, or inside the binary).

The binary pulsar Roemer delays comprise up to five **Keplerian parameters** describing elliptical orbits:

- the projected semimajor axis $x = (a \sin i)/c$
- longitude of periastron (ω) (the angle between a reference point and the point where the distance between the two stars of the binary are the closest.)
- time of periastron passage (T_0) (exact time when the binary system has the stars at least distance.)
- orbital period (P_b)
- orbital eccentricity (e)

Relativistic binaries, particularly those with compact and elliptical orbits, may allow the measurement of up to five **post-Keplerian (PK) parameters**.

Table 1 | Physical parameters for PSR J1614-2230

Parameter	Value
Ecliptic longitude (λ)	245.78827556(5) $^\circ$
Ecliptic latitude (β)	$-1.256744(2)^\circ$
Proper motion in λ	9.79(7) mas yr $^{-1}$
Proper motion in β	$-30(3)$ mas yr $^{-1}$
Parallax	0.5(6) mas
Pulsar spin period	3.1508076534271(6) ms
Period derivative	$9.6216(9) \times 10^{-21}$ s s $^{-1}$
Reference epoch (MJD)	53,600
Dispersion measure*	34.4865 pc cm $^{-3}$
Orbital period	8.6866194196(2) d
Projected semimajor axis	11.2911975(2) light s
First Laplace parameter (esin ω)	$1.1(3) \times 10^{-7}$
Second Laplace parameter (ecos ω)	$-1.29(3) \times 10^{-6}$
Companion mass	0.500(6) $M_\odot$
Sine of inclination angle	0.999894(5)
Epoch of ascending node (MJD)	52,331.1701098(3)
Span of timing data (MJD)	52,469–55,330
Number of TOAs†	2,206 (454, 1,752)
Root mean squared TOA residual	1.1 μ s
Right ascension (J2000)	16 h 14 min 36.5051(5) s
Declination (J2000)	$-22^\circ 30' 31.081(7)''$
Orbital eccentricity (e)	$1.30(4) \times 10^{-6}$
Inclination angle	$89.17(2)^\circ$
Pulsar mass	1.97(4) $M_\odot$
Dispersion-derived distance‡	1.2 kpc
Parallax distance	>0.9 kpc
Surface magnetic field	1.8×10^8 G
Characteristic age	5.2 Gyr
Spin-down luminosity	1.2×10^{34} erg s $^{-1}$

Figure 3.1: Physical parameters for PSR J1614-2239 (The numbers in parentheses for each parameter are the errors in the last digits of those parameters)

3.2 Pulsar Timing Arrays (PTAs)

The double neutron-star systems have indirectly confirmed the existence of gravitational waves by matching the predicted orbital decays as orbital energy is carried away by gravitational radiation. The key focus of pulsar astronomy has shifted towards the **detection of Gravitational Waves using Pulsar Timing Arrays (PTAs)**.

We have previously seen how stable the Millisecond Pulsars (MSP's) are. They act like precise clocks in space. Normally, an array refers to a collection of several telescopes that work together. But the PTA technique focused on a collection of multiple pulsars (rather than multiple instruments).

How does the technique work?

The time of arrival for a pulsar can be calculated theoretically. But the actual time varies. The gap is called **timing residual**.

$$\text{Residual} = \text{Observed} - \text{Predicted}$$

The most likely sources of detectable nanohertz GWs are supermassive black hole binaries (with total masses of $10^8 - 10^{10} M_{\odot}$).

In General Relativity, the gravitational waves are produced by **quadrupole changes** in mass distribution. By quadrupolar, we mean that the space is stretched in one direction and compressed in the perpendicular direction. Each pulsar sends regular pulses. In the presence of the GW's, there is a shift in pulse arrival time, timing residual.

If two pulsars are close together, the GW stretches space similarly along both lines of sight, their timing shifts are similar. This corresponds to positive correlation.

If pulsars are far apart, the GW stretches the space in one direction, and compresses in the other. Their timing shifts can be uneven (zero correlation), or even opposite (negative correlation).

Say we take a collection of pulsars. Then compare each pair, and plot the correlation value against the angle between them. What we get is the **Hellings–Downs curve**.

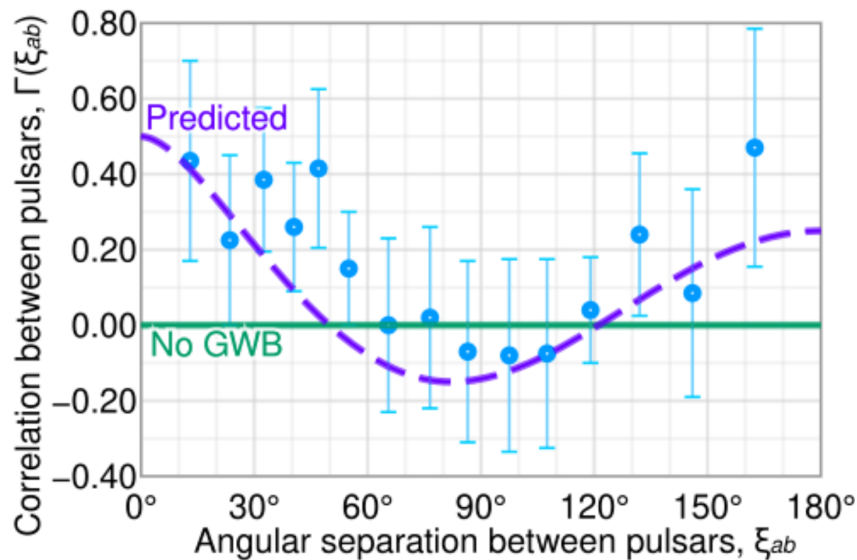


Figure 3.2: Hellings-Downs curve

3.3 Determination of Pulse Times of Arrivals

A paper by J. H. Taylor, titled "**Pulsar Timing and Relativistic Gravity**" (1992), provides a comprehensive review of pulsar timing techniques. TOA's are derived from averaging weak signals into profiles.

Let's say,

$p(t)$ = observed pulse signal

$s(t)$ = standard/template pulse

We sample (take measurements) at regular time intervals Δt .

So,

$$p_j = p(t_j); t_j = j\Delta t \quad (3.4)$$

$$p_0 = p(t_0); t_0 = 0$$

$$p_1 = p(t_1); t_1 = j\Delta t$$

$$p_2 = p(t_2); t_2 = 2j\Delta t$$

and so on.

Before sampling, we filter out the frequency range such that $f \leq \frac{1}{2\Delta T}$.
(where T is the emission time)

This is done to ensure that only the relevant signals are reached. To avoid filtering out useful data, Δt is taken very small.

If the above criteria are met, the **finite sampling theorem** ensures that the entire continuous signal is fully determined by its discrete samples. No information is lost. This is exactly the sampling theorem by **Nyquist-Shannon**; tailored for real, finite length data by Bracewell, 1965.

If $p(t)$ is equal to shifted and scaled replica of $s(t)$ plus some random noise, we can write

$$p(t) = a + b s(t - \tau) + noise$$

$p(t)$: observed pulse

$s(t)$: standard/template pulse

a : constant offset

b : scaling factor

τ : time shift

Importantly, a shift in time becomes a shift in phase in Fourier space. A shift in time changes the phase of frequency.

Specifically,

$$s(t - \tau) = e^{(i\omega t - i\omega\tau)} = e^{i\omega t} \cdot e^{-i\omega\tau}$$

where ω is a constant, and τ is the time offset.

The key identity is that

$$\text{Time shift} \implies \text{Multiply by } e^{-i\omega\tau}$$

It is preferable to deal in phase rather than in time (since phase can be treated as an **angle of a phasor diagram**). Take the Fourier transform of the observed signal. Then take the Fourier transform of the template. Compare phases (as angles of phasor).

We use the **Discrete Fourier Transform (DFT)** to convert the continuous real-world signals to discrete signals, which we are able to deal with.

Why?

Same reason, it is preferable to deal in phase rather than time. Discrete Fourier Transforms of the two profiles can be written as...

$$P_k \exp(i\theta_k) = \sum_{j=0}^{N-1} p_j \exp\left(i 2\pi j \frac{k}{N}\right) \quad (3.5)$$

$$S_k \exp(i\theta_k) = \sum_{j=0}^{N-1} s_j \exp\left(i 2\pi j \frac{k}{N}\right) \quad (3.6)$$

where P_k and S_k are the amplitudes of the complex Fourier coefficients (k is the frequency index), and θ_k and ϕ_k are the phases.

We write the combined equation as...

$$P_k \exp(i\theta_k) = aN + bS_k \exp[i(\phi_k + k\tau)] + G_k, \quad k = 0, 1, \dots, (N-1) \quad (3.7)$$

where the G_k term represents random noise. The bias a and scale factor b assume similar roles in the time and frequency domains.

After the transforms have been computed, the value of a can be directly obtained by putting $k = 0$, i.e.

$$a = \frac{P_o - bS_o}{N}$$

The desired pulse time of arrival τ as well as the gain factor ' b ' can be obtained by goodness-of-fit statistics.

What is goodness-of-fit statistics?

It is the measure of how well the model matches the observer data. For example, here we try to match the observed pulse signal ($p(t)$) and the standard pulse ($s(t)$). The goodness-of-fit (χ^2) is given by:

$$\chi^2 = \sum \left(\frac{\text{difference}}{\text{uncertainty}} \right)^2 \quad \text{or} \quad \left(\frac{\text{noise}}{\text{uncertainty}} \right)^2 \quad (3.8)$$

Here,

$$\chi = \sum_{k=1}^{N/2} \left| \frac{P_k - bS_k \exp[i(\phi_k - \theta_k + k\tau)]}{\sigma_k} \right|^2$$

The additional term σ_k here is the root-mean-square amplitude of the noise at freq. k . This model tries to minimize this χ^2 . Smaller the χ^2 value, better the result. We try to get these a , b , τ such that the model best matches the data.

Lets understand why the limits are $k = 1$ to $N/2$. For that, we need to understand the symmetry.

For any real signal, $P_{N-k} = P_k^*$.

This is called conjugate/Hermitan symmetry (the $*$ represents the complex conjugate).

These are called the mirror copies of each other (same amplitude, opposite phase). The indices are equal in a way as:

$$k = N - k$$

$$1 = N - 1$$

$$2 = N - 2$$

(and so on...)

So we simplify the summation term.

$$\sum_{k=0}^{N-1} (...) = 2 \sum_{k=1}^{N/2} (...)$$

But the extra factor 2 is not required to be multiplied since each symmetric term represents both components.

From equation (3.7)

$$\begin{aligned} P_k e^{i\theta_k} &= b S_k e^{i(\phi_k + k\tau)} + G_k \\ P_k e^{i\theta_k} - b S_k e^{i(\phi_k + k\tau)} &= G_k \end{aligned}$$

This is now rearranged as data - model = noise.

From equation (3.8),

$$\chi^2 = \sum_k \frac{|P_k e^{i\theta_k} - b S_k e^{i(\phi_k - \theta_k + k\tau)}|^2}{\sigma_k^2}$$

But $e^{i\theta} = 1$, so

$$\chi^2 = \sum_k \frac{|P_k - b S_k e^{i(\phi_k - \theta_k + k\tau)}|^2}{\sigma_k^2} \quad (3.9)$$

Expand the Numerator:

$$\begin{aligned} |A - B|^2 &= (A - B)(A - B)^* \\ |P_k - b S_k e^{i\alpha_k}|^2 &= [P_k - b S_k e^{i\alpha_k}] [P_k - b S_k e^{-i\alpha_k}] \end{aligned}$$

(where $\alpha_k = \phi_k - \theta_k + k\tau$)

$$|P_k - b S_k e^{i\alpha_k}|^2 = P_k^2 - b P_k S_k e^{i\alpha_k} - b P_k S_k e^{-i\alpha_k} + b^2 S_k^2$$

But $e^{i\alpha_k} + e^{-i\alpha_k} = 2 \cos \alpha_k$

So,

$$|P_k - b S_k e^{i\alpha_k}|^2 = P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos \alpha_k$$

Substitute this back into eq (3.9)

$$\begin{aligned} \chi &= \sum_k \frac{P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos \alpha_k}{\sigma_k^2} \\ \chi &= \sum_k \frac{P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos(\phi_k - \theta_k + k\tau)}{\sigma_k^2} \end{aligned} \quad (3.10)$$

A point to note here is that χ^2 is exactly the goodness-of-fit statistic here. The quantity does not precisely be in some range (like probability has a range 0 to 1). χ^2 **is a function of b and τ** . We do not care about the absolute χ^2 . We care about the value of τ that **minimises** χ^2 .

At the global minimum, the derivative of χ^2 wrt τ and b must be zero. Refer eq (3.10).

$$\frac{\partial \chi^2}{\partial \tau} = \frac{2b}{\sigma^2} \sum k P_k S_k \sin(\phi_k - \theta_k + k\tau) = 0 \quad (3.11)$$

$$\frac{\partial \chi^2}{\partial b} = \frac{2b}{\sigma^2} \sum S_k^2 - \frac{2}{\sigma^2} \sum P_k S_k \cos(\phi_k - \theta_k + k\tau) = 0 \quad (3.12)$$

Solving,

$$b = \frac{\sum P_k S_k \cos(\phi_k - \theta_k + k\tau)}{\sum S_k^2}$$

Any smooth function near a minimum behaves like a quadratic (parabola).

By Taylor expansions:

$$f(x) = f(x_0) + \frac{1}{2} f''(x_0) (x - x_0)^2$$

For χ^2 :

$$\begin{aligned} \chi^2 &= \chi_{min}^2 + (curvature)(\tau - \tau_0^2) \\ \chi^2(\tau) &= \chi_{min}^2 + \frac{1}{2} \frac{\partial^2 \chi^2}{\partial \tau^2} (\tau - \tau_0^2) \end{aligned} \quad (3.13)$$

(This is the Taylor expansion around the best-fit value)

By definition of uncertainty (σ_τ), it is the shift in τ that increases χ^2 by 1.

$$\chi^2 = \chi_{min}^2 + 1$$

From eq (3.13)

$$\begin{aligned} 1 &= \frac{1}{2} \frac{\partial^2 \chi^2}{\partial \tau^2} (\sigma_\tau^2) \\ \sigma_\tau^2 &= 2 \left(\frac{\partial^2 \chi^2}{\partial \tau^2} \right)^{-1} \end{aligned}$$

The factor of 2 is "absorbed" because the summation is redefined to run only over independent frequency pairs.

So,

$$\begin{aligned} \sigma_\tau^2 &= \left(\frac{\partial^2 \chi^2}{\partial \tau^2} \right)^{-1} \\ \sigma_\tau^2 &= \frac{\sigma^2}{2b \sum k^2 P_k S_k \cos(\phi_k - \theta_k + k\tau)} \\ \sigma_b^2 &= \frac{\sigma}{2 \sum S_k^2} \end{aligned}$$